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Impact of Lifetime on U.S. Residential Building LCA Results

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Abstract

Purpose Many life cycle assessment (LCA) studies do not adequately address the actual lifetime of buildings and building products, but rather assume a typical value. The goal of this study was to determine the impact of lifetime on residential building LCA results. Including accurate lifetime data into LCA allows a better understanding of a product's environmental impact that would ultimately enhance the accuracy of LCA results.

Methods This study focuses on refining the U.S. residential building lifetime, as well as lifetime of interior renovation products that are commonly used as interior finishes in homes, to improve LCA results. Residential building lifetime data that presents existing trends in the U.S. was analyzed as part of the study. Existing data on product emissions were synthesized to form statistical distributions that were used instead of deterministic values. Product emissions data were used to calculate life cycle impacts of a residential model that was based on median U.S. residential home size. Results were compared to existing residential building LCA literature to determine the impact of using updated, statistical lifetime data. A Monte Carlo analysis was performed for uncertainty analysis. Sensitivity analysis results were used to identify hotspots within the LCA results.

Results and discussion Statistical analysis of U.S. residential building lifetime data indicate that average building lifetime is 61 years and has a linearly increasing trend. Interior renovation energy consumption of the residential model that was developed by using average U.S. conditions was found to have a mean of 220 GJ over the life cycle of the model. Ratio of interior renovation energy consumption to pre-use energy consumption, which includes embodied energy of materials, construction activities, and associated transportation was calculated to have a mean of 34% for regular homes and 22% for low-energy homes. Ratio of interior renovation to life cycle energy consumption of residential buildings was calculated to have a mean of 3.9% for regular homes and 7.6% for low-energy homes.

Conclusions Choosing an arbitrary lifetime for buildings and interior finishes, or excluding interior renovation impacts introduces a noteworthy amount of error into residential building LCA, especially as the relative importance of materials use increases due to growing number of low-energy buildings that have lower use phase impacts.

Keywords Lifetime, Interior renovation, Residential buildings, Environmental life cycle assessment, Uncertainty analysis, Monte Carlo analysis

1 Introduction

The built environment is a major contributor to both social and economic development and represents a large portion of real capital in many countries; but it is also a primary source of environmental impacts.

Furthermore, existing building stock requires continuous investments for repair and renovations [1]. Of the 2.5 billion metric tons of non-fuel materials that moved through the economy in 1990, over 70% were used for construction [2]. In 2010, buildings are estimated to account for close to 40% of U.S. primary energy consumption and greenhouse gas emissions [3].

The notion that building structures that would last for centuries is the best environmental solution to our problems does not match with our existing building use trends and knowledge of the built environment. Buildings will be replaced with newer designs that are more suited towards the needs of future occupants. This concept should be considered during initial design, construction, and environmental and economic analysis. History of Kingdome in Seattle, Washington, illustrates the importance of including anticipated lifetime of structures during decision-making. Being the largest concrete dome of its time, the stadium was built in 1976 for \$67 million with a design lifetime of 75 years, extending potentially up to 120 years with scheduled maintenance. However, the stadium was closed for major repair in 1994 after several ceiling tiles fell, and cost \$70 million in repair work. The structure was demolished 6 years after repairs only to be replaced with a new stadium that had cost \$430 million when completed [2]. Not accounting for actual lifetime of buildings could have significant economic and environmental consequences.

In many cases, building lifetime is governed by factors not directly related to the building design. For residential buildings in the U.S., lifetime is more directly related to social acceptability factors rather than durability or structural problems [4].

Life cycle assessment (LCA) is a tool that can quantify the environmental impacts of buildings [5]. However, many building LCA studies do not adequately address the actual lifetime of residential buildings and building products, but rather assume a typical value, say 50 years [6-9]. This study addresses a gap by determining the impact of lifetime on residential building LCA results. Including accurate lifetime information into LCA allows a better understanding of the life cycle impacts, ultimately enhancing the accuracy of LCA studies.

This study focuses on refining the U.S. residential building lifetime, as well as lifetime of interior renovation products such as paint and carpet that are commonly used as interior finishes in homes, to improve LCA results. Residential building lifetime presented in this study is a new contribution to literature. Existing data on product emissions were synthesized to form statistical distributions that were used instead of deterministic values. Product emissions data are used to calculate life cycle impacts of a residential model that is based on median U.S. residential home size. Results were compared to existing residential building LCA literature to determine the impact of using updated, statistical lifetime data. A Monte Carlo analysis was performed for uncertainty analysis. Sensitivity analysis results were used to identify hotspots within the LCA results.

1.1 Building use and lifetime

Lifetimes of commercial and residential buildings are significantly different. Although some studies assume a 50 year life span for office buildings [10], reported lifetimes vary from 12-15 years [11] to 20 years [12, 13]. Furthermore, department stores undergo extensive interior renovations every 3-5 years for branding and marketing [12]. Although a quantitative analysis for residential building lifetimes has not been conducted, the general consensus is that residential buildings have longer lifetimes ranging from 50 to 100 years [4, 6-9, 12, 14-24]. However, building lifetimes used in LCA studies are often arbitrarily selected as explicitly stated in many studies [4, 6-9, 12, 14, 16, 18-20, 22, 24].

1.2 Factors that influence lifetime of building products

Reasons behind replacing interior finishes can be grouped into three categories: failure, dissatisfaction, or change in consumer needs [25]. Failure is related to durability of materials and is the only category that can be designed or influenced by the manufacturer. All materials degrade over time as they are used. In addition to normal wear and tear, UV light, humidity, temperature, biological factors, installation and maintenance procedures are key degradation factors that affect durability of interior building products.

Dissatisfaction is mostly associated with styling changes, fashion trends, or new products being introduced to the market. In this case, consumers are not necessarily motivated by rational cost-benefit considerations, but rather by their desires and perceptions. Occupant needs may change over time even at the same residence, when occupants have children or become elderly for instance.

In practice, the actual lifetimes of various building products are shorter than what they had been designed for [26, 27]. Occupant behavior influenced by societal trends is an important factor that influences the lifetime of products [28-30]. However, models that capture the effects of consumer behavior on product lifetimes are not widely used to the best of our knowledge. Lifetime estimation methods that can capture consumer behavior are a necessary step towards modeling lifetime [31].

2 Methods

Methods used to gather and process data, together with assumptions made and equations used to calculate results are described in this section. Data sources for residential building lifetime and interior finishes are presented. Multiple data points enabled the use of distributions for variables. Procedure used to fit distributions, and uncertainty analysis of results using the Monte Carlo method are described. Results were applied to a residential model for interpretation. A description of the residential model is presented together with related assumptions. Data on different life cycle phases of a residential building were also analyzed in order to compare interior renovation impacts to life cycle impacts.

2.1 Data sources

Multiple data sources were used to determine the lifetimes for this study. Data published by the U.S. Census Bureau were used extensively for building related statistics [32-42]. BEES v4.0 [43], and the Ecoinvent v2 and ETH-ESU LCI databases incorporated in Simapro v7.1 software [44] provided the majority of environmental emissions data for building products. Traci 2 v3.01 [45] was used for impact assessment of inventory data.

2.1.1 Residential Building lifetime

Accurate data on residential building lifetime was vital since building lifetime determines the number of interior renovations. Data on U.S. residential building stock was published by the U.S. Census Bureau under the 2009 American Housing Survey microdata, which had a sample size of over 70,000 residences [38]. No other governmental or public source provided such a large number of reliable data points on the U.S. housing stock. Survey microdata included data for when a building was built and whether it was demolished since the last survey. The difference between these two values provided the lifetime for that building. A large dataset including over 3,700 data points for building lifetime was gathered from microdata by this approach.

A caveat of using this data source was that the type of building was not recorded for buildings that were coded as demolished. Therefore, average lifetime of different building types could not be calculated

directly from this primary source. However, it was possible to reach a conclusion regarding average lifetime of single-family residential buildings based on three supporting analyses.

On a national scale, single-family detached houses and apartments form 63% and 25% of the U.S. building stock, respectively [39]. The remaining portion being equally divided between single-family attached homes and mobile units. The difference in average building lifetime of single-family detached houses and apartments was investigated. Average age of existing single-family detached houses and apartments including two or more units were calculated to be 42.4 years and 44.1 years respectively from the 2009 American Housing Survey microdata [38]. Average age of existing buildings is different from building lifetime, since the building needs to be demolished in order to calculate its lifetime. Nevertheless, the difference in mean age of existing buildings between these two categories was found to be insignificant compared to the inherent uncertainty of building age.

Existing buildings were separated according to type and year built. The ratio of single-family detached houses to single-family detached houses and apartments varies within a range of 60-80% over the decades but has an almost constant trend at 70%. Therefore, no evidence was found to support that single-family detached houses and apartments have different lifetimes, and so they were assumed to be the same throughout the current study. A study by O'Connor surveying 227 demolished buildings found that only 8 were demolished due to structural reasons, and that buildings were usually demolished due to changing land values and occupant needs [46]. Results of this study support our assumptions since social factors independent of building type were found to determine building lifetime in most cases.

Buildings built prior to 1920, which constitute 7% of the existing U.S. building stock, were presented in a single category in the 2009 American Housing Survey results [38]. The 2008 New York Housing Survey divides this category into two sections: structures built between 1900-1919, and those built pre-1900, with ratios of 75% and 25% respectively [42]. The same ratios of 75% and 25% were used to further classify pre-1920 buildings on a national basis into two separate categories of 1900-1919, and pre-1900.

The methods described here were used on past surveys as well to observe the trend in residential building lifetime. Survey results dating back to 1997 were published by the U.S. Census Bureau and were used in this study to plot trends in residential building lifetime [32-38].

2.1.2 Products investigated

Interior finish products that are commonly replaced within U.S. residential buildings were investigated in this study. Paint is usually applied in all buildings to some degree, and therefore was included. Multiple flooring alternatives including carpet, hardwood, linoleum, vinyl, and ceramic were also considered.

Data points for lifetime and environmental emissions of interior finishes that were used in the study are given in Table 1. In some instances, a range of values was provided for lifetime of products rather than a single value [47, 48]. In these cases, a uniform distribution was assumed for the given range of values. For long lasting products such as hardwood and ceramic, some sources indicated that the product was expected to last as long as the building, therefore not necessitating any interior renovation [47, 48]. Due to large uncertainty associated with predicting product lifetime for several decades into the future, the lower lifetime limit was selected during analysis, i.e. 75 years when lifetime was given as 75 or more years.

Table 1 Data points for lifetime and environmental emissions of interior finishes

	Paint	Carpet	Hardwood	Linoleum	Vinyl	Ceramic
Lifetime (years)	3 [49], 4 [43], 5 [50], 51], 7 [52], 8 [21], 10 [6, 7, 22, 53]	5 [24, 47], 8 [6, 54], 8-10 [48], 9 [55], 10 [52], 11, 15 [43], 12 [21, 51], 17 [7]	10 [17], 20 [17, 24], 25 [17], 40 [56, 57], 45 [23, 55], 50 [7, 17, 21], 50+ [47], 100+ [48]	7-40 [47], 15 [54, 55], 20 [58, 59], 23 [55], 25 [48, 56, 57], 30 [43]	7-40 [47], 8 [54], 9, 23 [55], 17 [21], 18 [51], 20 [6, 52, 56, 57, 59, 60], 40 [43], 50 [48]	20 [24, 61], 30 [21], 50 [43], 75 [51], 75-100 [48]
Energy (MJ/m ²)	3.0, 3.7, 7.2 [43], 3.6 [7], 6.6, 6.7, 6.8 [44], 11 [6],	89, 102, 111, 122, 131, 209, 214, 239, 242, 242, 253, 274, 276, 282, 285, 296, 320 [43], 171 [62], 183 [63]	250 [47], 314, 402, 582 [23], 530, 530, 550, 920 [17]	57.7 [57], 130 [47], 161 [59], 276, 305 [43]	56 [57], 130 ^a [51], 165 [47], 170 [59], 245 [43]	347 [43]
Global Warming Potential (kg CO ₂ E/m ²)	0.05, 0.09, 0.18 [43], 0.26, 0.27, 0.37, 0.38 [44]	5, 5, 10, 11, 12, 12, 12, 12, 13, 13, 15, 17 [43], 10.6 [62], 11.3 [63]	4.4, 5.9, 7.1, 12.7 [17], 29 [47], 44, 56, 56 [23]	1.6 [57], 2.6 [54], 6, 10 [43], 17 [47]	4.1 [57], 9.4 [54], 12 [47], 10 [43]	23 [44], 26 [43]
Acidification (g H ⁺ /m ²)	0.03, 0.04, 0.08 [43], 0.06, 0.09, 0.12, 0.15 [44]	2, 2, 2, 2, 3, 4, 5, 5, 5, 5, 5, 5, 6, 8, 8 [43], 2.1 [63], 2.5 [62]	5200, 5400, 5700, 11300 [17] ^b , 5100, 6100, 6600 [23] ^b	1.2 [47], 5.6, 6 [43]	2 [47], 6 [43]	4.3 [44], 9.6 [43]
Eutrophication (g N/m ²)	0.00 [43], 0.03, 0.53, 0.96, 1.29 [44]	2, 2, 2, 2, 3, 3, 4, 5, 10, 11, 12, 13, 13, 14, 15 [43], 10 [62], 12 [63]	2, 31, 38 [23], 35, 35, 38, 81 [17]	18.9, 23.3 [43]	0.02 [57], 1.7 [43]	4 [43], 8.3 [44]
Smog (g NO _x /m ²)	0.5, 0.6, 1.0, 1.1 [44], 16.2, 16.5, 16.9 [43]	24, 24, 24, 25, 28, 33, 47, 50, 58, 58, 61, 63, 64, 64, 64, 64 [43]	-	119, 125 [43]	40 [43]	38 [44], 122 [43]

Notes: The sign '+' after a number indicates that expected lifetime was more than the given value.

Multiple references after a data point indicate multiple occurrences in different studies.

^a Average values were used to convert mass to volume

^b TRACI characterization factors were used to convert SO₂ into g H⁺

2.2 Uncertainty in variables

Addressing uncertainty plays a key role in interpreting results of life cycle studies. The use of distributions for lifetime and environmental emissions data was preferred over using deterministic values since a realistic uncertainty analysis was not possible otherwise. @Risk v5.5 was used for uncertainty analysis [64].

The chi-squared test was used to fit distributions. A goodness-of-fit test is an inferential procedure used to determine how well a given set of data fits a chosen distribution [65]. Originally developed by Pearson in 1900, the chi-squared test is the oldest inference procedure that is still used today in its original form [66]. A Weibull distribution provided the best fit for residential building lifetime and lifetime of most interior finishes. The use of this distribution to model lifetime is common, supported by standards and guidelines [67-69].

Developments in the field of building LCA's have not been matched by accurate emissions data for building products [70]. Existing databases do not sufficiently cover the vast array of products that exist today. Few data points were located for some building products' environmental impact categories due to lack of reliable publications and confidentiality concerns from the manufacturer's perspective. Therefore, triangular or uniform distributions were defined for variables where an adequate number of data points could not be found.

Since variables were defined as distributions, interior renovation impact results were also calculated as distributions having a mean and a confidence interval. Monte Carlo simulation was used to calculate uncertainty in results. Monte Carlo is a statistical method that uses random values from input parameters and presents a distribution for the output parameter [71, 72]. The likelihood of potential outcomes can thus be observed from resulting distributions; 20,000 iterations were used for analysis in this study.

2.3 Interpretation of results through a developed residential model

The goal of the study was to determine the impact of lifetime on residential building LCA. Interior renovation impacts over the life cycle of a residential building model were calculated by using the determined distributions for building lifetime, building product lifetime, and environmental impact of products. A residential model based on median U.S. residential building size was used to calculate life cycle environmental impacts of interior renovation. Existing single-family detached homes have a median size of 167 m² based on the 2009 American Housing Survey microdata, which was used to determine the size of the residential model in this study [38]. The mean single family detached home size of 206 m² for existing residential buildings calculated from the same microdata places more emphasis on larger homes as compared to the distribution of home size and therefore was not preferred.

A 4-bedroom, 2-bathroom home was assumed for the residential model with the following specifications: Ceiling and interior walls were painted, bathroom walls were painted up to half height and the remaining portion covered with ceramic, wall to wall carpeting for the home except for kitchen where vinyl covering was assumed. In total, 550 m² of painted surface area, 45 m² of ceramic, 122 m² of carpeting, and 21 m² of vinyl were calculated for the residential model. The interior painted surface area was highly dependent on design, or architectural model of the home, and so a uniform distribution of 500-600 m² was used during calculations to account for the high level of uncertainty associated with painted surface area.

Equation 1 was used to calculate energy use and environmental emissions of interior finishes over the life cycle of a building. The given equation was used to calculate interior renovation impacts and does not include initial construction stage material use. A 5% waste factor was assumed for all floor-covering materials as construction loss from cutting and fitting of products. This value was based on manufacturer recommendations and examples of its use exist in literature [6]. The same type of product as the previous layer was assumed to be used during interior renovation (e.g. carpet replaced with carpet) throughout the lifetime of the building.

$$\left[\frac{\text{building lifetime (years)}}{\text{product lifetime (years)}} - 1 \right] \times \frac{\text{product emissions (kg CO}_2\text{E/m}^2\text{)}}{\text{efficiency (m}^2\text{/m}^2\text{)}} \times \text{application area (m}^2\text{)} \quad (1)$$

2.4 Residential building energy consumption over different life cycle phases

Calculating interior renovation impacts of the residential model enables comparisons to be made between different life cycle phases of a residential building. A distinction can also be made between residential buildings built by using regular materials and techniques, and that are designed to consume less energy during their use phase, or low-energy homes. Consuming less energy during use phase, which is the dominating phase for regular homes, increases the relative importance of other life cycle phases including interior renovation.

Pre-use phase, which includes initial materials use, construction, and associated transportation for both activities, has a mean energy consumption of 4.0 GJ/m² with a range of 1.7-7.3 GJ/m² based on results of multiple case studies on residential buildings [4, 6, 8, 15, 16, 21, 73]. Pre-use energy consumption of low-energy homes was found to have a higher mean of 6.2 GJ/m² with a range of 4.3-7.7 GJ/m² [6, 8, 9]. A contributing factor for increased energy intensity in low-energy homes is the thicker shell and the high embodied energy associated with insulation products that are applied for weatherization.

Mean energy consumption during the use phase of existing single-family detached homes in the U.S. is given by the Energy Information Administration to be 0.45 GJ/m²/yr [74]. A separate category for low-energy buildings was not present in this primary source. Use phase energy consumption of low-energy homes was estimated to be 0.18 GJ/m²/yr with a range of 0.07-0.41 GJ/m²/yr from published case studies [6, 8, 9].

Demolition energy and transportation of waste was found to be 0.1-1% of life cycle energy regardless of building type and so was neglected during calculations [4, 6, 51, 75].

Total energy consumption over the life cycle of the residential model can then be modeled by using Equation 2.

$$[\text{pre - use (GJ/m}^2) \times \text{area (m}^2)] + [\text{use (GJ/m}^2/\text{yr)} \times \text{area (m}^2) \times \text{building lifetime (yrs)}] \quad (2)$$

2.5 Sensitivity analysis and validation

Energy consumption and environmental emissions of products over the residential buildings life cycle are a function of multiple variables including multiple products. After a Monte Carlo simulation was performed for an environmental impact category, a sensitivity analysis was conducted to identify variables that contributed most to interior renovation impacts of the residential model.

Paint and flooring alternatives were assumed not to influence residential building lifetime and were also assumed not to affect renovation cycles of other products included in the analysis. This enabled the use of independent variables in the sensitivity analysis.

Findings of this study were applied to published case studies to compare results. A journal article on residential building LCA that is frequently cited by other researchers was chosen to validate results. The applicability of research findings and the level of detail that was presented in the article were also considered during selection.

3 Results and Discussion

3.1 Residential building lifetime

Average residential building lifetime was calculated to be 61 years with a standard deviation of 25 years based on the 2009 American Housing Survey. Lifetime is expected to be within a large range of 21 years to 105 years with 90% confidence. Weibull distribution with a shape parameter of 2.8 and a scale parameter of 73.5 provided the best fit to model lifetime of residential buildings. By using the same method, residential building lifetime was also calculated from previous surveys. Figure 1 presents lifetime distribution results for housing surveys conducted from 1997 to 2009.

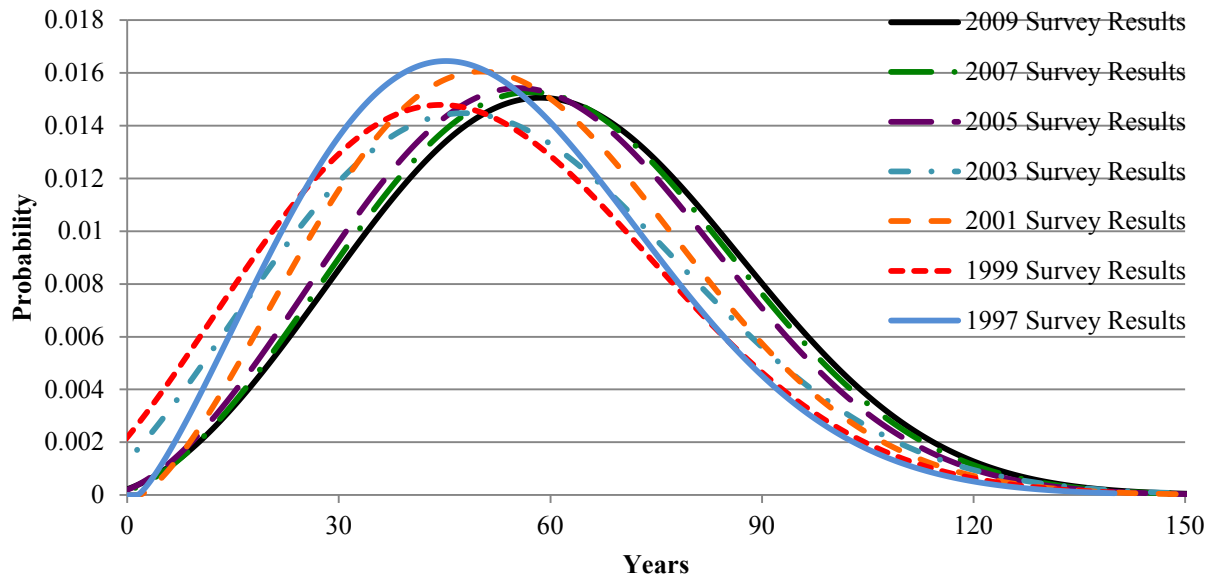


Figure 1 Lifetime distribution of residential buildings calculated from multiple American Housing Survey microdata

3.2 Product lifetimes and environmental emissions

Table 2 presents the mean and coefficient of variation values for lifetime, energy consumption, and environmental emissions data for each product. The coefficient of variation is defined as the ratio of the standard deviation of a distribution to its mean, and is a measure of dispersion in data. Table 2 also presents the type of distribution used for each variable, which were developed as described in section 2.2.

Table 2 Mean values, coefficient of variation, and the type of distribution for each variable that is used in this study

	Paint	Carpet	Hardwood	Linoleum	Vinyl	Ceramic
Lifetime (years)	6.9 (0.39, w)	10 (0.32, w)	42 (0.52, w)	22 (0.19, w)	22 (0.45, w)	48 (0.45, t)
Energy (MJ/m ²)	6.8 (0.41, t)	220 (0.31, t)	570 (0.44, u)	200 (0.40, t)	160 (0.41, t)	350 (0.08, u)
Global Warming Potential (kg CO ₂ E/m ²)	0.2 (0.32, t)	11 (0.28, t)	38 (0.33, t)	10 (0.58, u)	9.3 (0.31, t)	25 (0.12, u)
Acidification (g H ⁺ /m ²)	0.1 (0.41, t)	5.0 (0.33, t)	6,300 (0.23, t)	4.5 (0.41, t)	4.0 (0.41, t)	7.0 (0.25, u)

Eutrophication (g N/m ²)	0.8 (0.58, u)	8.5 (0.50, u)	62 (0.52, t)	21.0 (0.14, u)	1.0 (0.58, u)	6.0 (0.58, u)
Smog (g NO _x /m ²)	17 (0.02, u)	50 (0.23, t)	-	120 (0.05, u)	40 (0.14, u)	80 (0.58, u)

Note: Numbers in parentheses represent the coefficient of variation for that distribution. Letters that follow denote the type of distribution used, where, w=Weibull, u=Uniform, t=Triangular.

3.3 Environmental emissions of the residential model

Impacts of interior renovation over the life cycle of a residential building were quantified. Table 3 presents the energy consumption and environmental emissions of products that were applied to the residential model. Results from all products used in the residential model are combined for each impact category and presented together with a range of results with a 90% confidence interval and the associated standard deviation for the resulting distribution. The combined results represent interior renovation impacts throughout the lifetime of the residential model.

Table 3 Environmental impacts of interior renovation for the residential model

	Lower Bound	Mean	Upper Bound	Coefficient of variation
Energy (GJ)	38	220	500	0.64
Global Warming Potential (t CO ₂ E)	1.9	11	24	0.63
Acidification (kg H ⁺)	0.8	4.6	11	0.67
Eutrophication (kg N)	1.7	10	24	0.71
Smog (kg NO _x)	28	130	270	0.57

Note: Lower and upper boundaries are for a 90% confidence interval

3.4 Comparing interior renovation to different life cycle phases of the residential model

Environmental impact results of interior renovation over the life cycle of the residential model was used to compare different life cycle phases of a residential building. Energy consumption of interior renovation compared to pre-use phase energy consumption was calculated to have a mean of 34% for regular homes, and 22% for low-energy homes. Figure 2 shows distribution of results together with ranges for the 90% confidence interval. The ratio of interior renovation energy to life cycle energy of residential buildings was found to have a mean of 3.9% for regular homes and 7.6% for low-energy homes. Figure 3 shows distribution of results together with a 90% confidence interval.

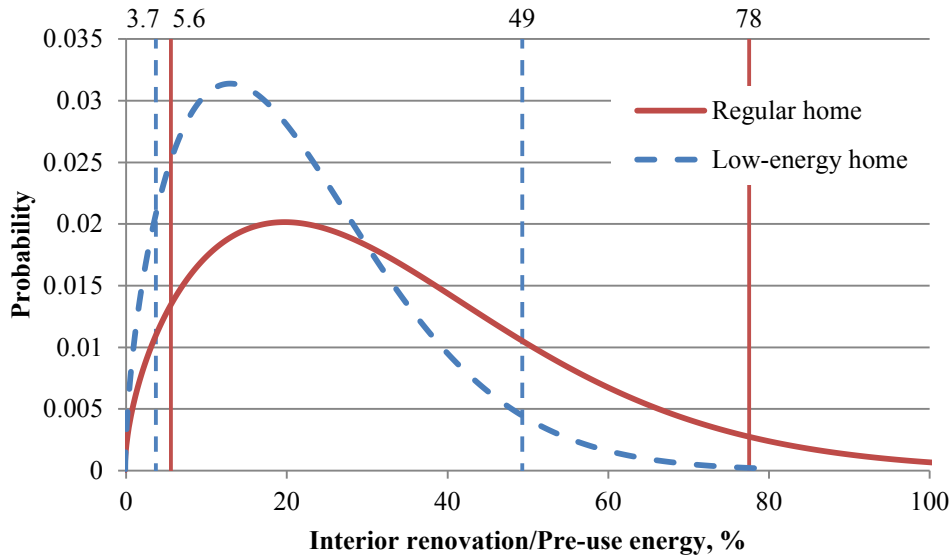


Figure 2 Distribution for the ratio of interior renovation energy to pre-use phase energy. Given error bars are for a 90% confidence interval

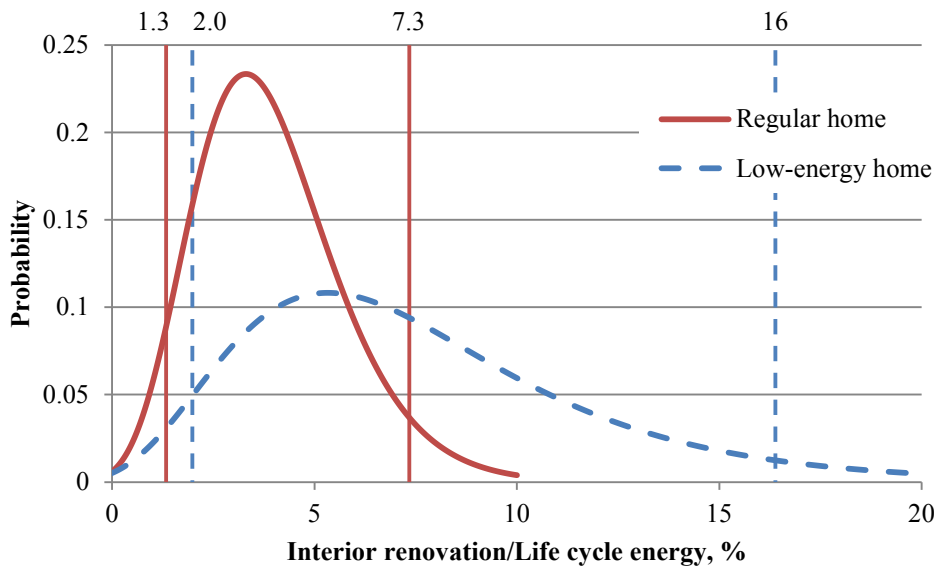


Figure 3 Distribution for the ratio of interior renovation energy to life cycle energy. Given error bars are for a 90% confidence interval

Techniques and materials that improve energy efficiency during the use phase of a building exist today, and are being increasingly applied to new residential constructions. The rapid increase in the number of low-energy buildings signifies public interest towards efficiency and preservation, which will further drive building efficiencies higher. As buildings become more efficient, their use phase emissions will decrease, which will increase the relative importance of interior renovation over the life cycle of a building.

3.5 Sensitivity analysis

Sensitivity analysis was performed to determine which variables had the greatest impact on results from the residential model. Figure 4 presents results for energy consumption analysis. A positive regression

coefficient indicates that results are directly proportional with changes in that category, whereas negative values indicate an inverse trend. A higher magnitude for the coefficient implies greater impact of that variable on results.

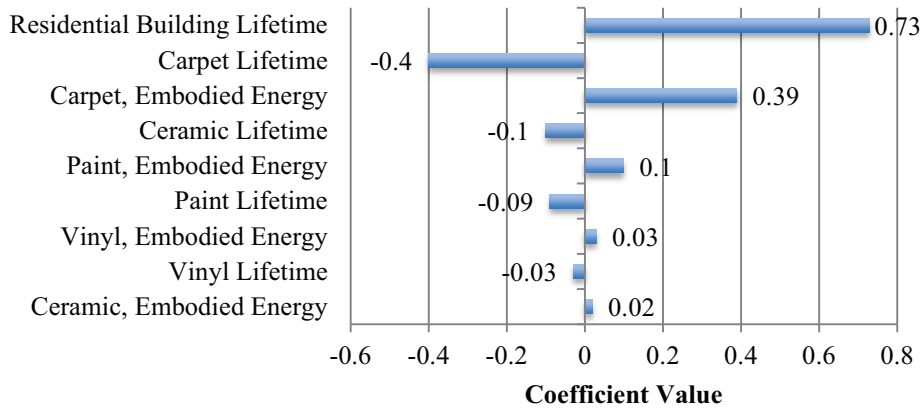


Figure 4 Sensitivity analysis results for energy consumption of the residential model, ranked based on decreasing influence to results

Results of sensitivity analyses should be used to identify hotspots and ultimately improve accuracy of LCA results. More accurate data should be sought for parameters having the greatest impact to improve accuracy of the study. Lifetime data and energy consumption of several interior renovation products were found to equally affect results for the residential model. Therefore, assuming an arbitrary lifetime for products would decrease accuracy as much as choosing a generic emissions factor for building products.

Residential building lifetime was found to have the greatest impact on interior renovation impacts. Following building lifetime, carpeting was found to have the most impact on results. Therefore, a recommendation for future LCA's involving similar materials and conditions would be to focus more on finding accurate data for carpeting compared to other interior finish products.

An additional sensitivity analysis was carried out on distribution selections since different distributions can be selected for a variable. As described in section 2.2, selections were based on chi-square test results for the fit between data and proposed distribution. In order to test the impact of distribution selection on end results, distributions different from the ones shown in Table 2 have been chosen, and results recalculated. However, results showed minimal variation in the statistical properties of variables.

3.6 Validation of results

A study by Keoleian et al. focused on life-cycle energy consumption of a 228 m² single-family house in the U.S. [6]. A 50-year residential building lifetime was assumed in the analysis. Renovation impacts have been presented in detail, which allowed results to be directly compared. A description of materials included in the study, together with assumed lifetime and embodied energy data were provided. Renovation cycles were set at 10, 8, and 20 years for paint, carpet, and vinyl respectively.

Interior renovation impacts have been revised by updating both residential building and building products lifetime. Energy consumption of interior renovation over the residential model lifetime was

found to be statistically the same; updated mean value of 370 GJ compared to 320 GJ estimated from figure given in the study. Although there is an increase of 15% in the calculated mean when results are revised for lifetime, the results were within the range of expected results given by the confidence interval. Since similar materials were used in both residential building analyses, the results are in support of each other. Revised energy consumption ratio of selected interior finishes compared to life cycle energy consumption of the model yield 2.3% and 5.8% for regular and low-energy homes respectively, which are also in accordance with results found in this study.

3.7 Trends in residential building lifetime

Each American Housing Survey contains information regarding the mean age of demolished buildings, and therefore a residential building lifetime trend was plotted by applying the procedure described in section 2.1.1 to past surveys. Results given in Figure 5 show an almost linear increasing trend in mean residential building lifetime in the last decade.

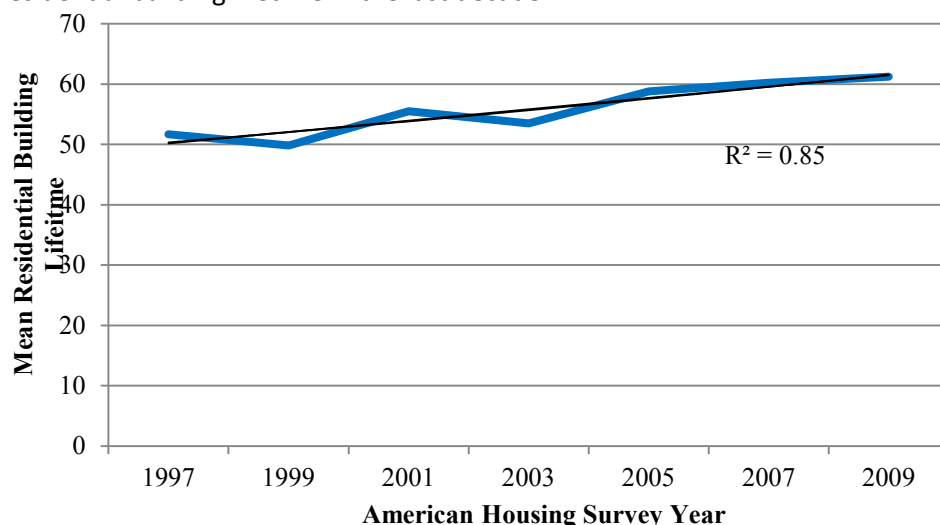


Figure 5 Change in U.S. residential building lifetime in the last decade [32-38, 76]

This outcome has important implications. Given that the increasing trend continues in the future, a recently built home would be expected to have a lifetime greater than 61 years. However, the observed linear increasing trend cannot continue indefinitely and there is expected to be an upper limit to achievable residential building lifetime dictated by structural design requirements or future technological improvements and demands. A model to predict trends in residential building lifetime is intended to be a future study.

Conclusions

Residential building lifetime data that presents existing trends in the U.S. was analyzed as part of the study. Results indicate that residential building lifetime in the U.S. is currently 61 years and has a linearly increasing trend. Existing LCA rely heavily on estimates for residential building lifetime, and choices are usually made arbitrarily. To our knowledge, this study is the first time mean residential building lifetime has been calculated from a large, reliable sample and used in LCA.

Lifetime of buildings and products presented in the current study should not be taken as static values. Future trends, occupant behavior, population demographics, regulatory policies, or development of new

technologies have the potential to alter both lifetime and emissions of buildings and building products. The increasing trend in residential building lifetime was demonstrated in the current study. Range of values supported by statistical analysis was used throughout the study to compensate for some of the uncertainties associated with variables. The use of distributions instead of deterministic values for lifetime of products and buildings improves accuracy of the study and makes results more objective. More data on environmental emissions of interior finishes is also a necessary step towards more robust results.

Interior renovation energy consumption for the residential model that was developed by using average U.S. conditions was found to have a mean of 220 GJ over the life cycle of the model. Using published data on energy consumption during pre-use and use phase of residential buildings enabled comparisons to be made among interior renovation impacts and other life cycle phases. Ratio of interior renovation to pre-use energy consumption was calculated to have a mean of 34% for regular homes and 22% for low-energy homes. Ratio of interior renovation to life cycle energy consumption of residential buildings was calculated to have a mean of 3.9% for regular homes and 7.6% for low-energy homes.

Life cycle impacts of traditional buildings are dominated by use phase emissions. However, this is likely to change as buildings become more energy efficient during their use phase. If the rapid increase in the number of low-energy buildings observed in the last decade continues into the future, use phase emissions of a building would decrease, increasing the relative importance of interior renovation in the life cycle of a residential building. Such an increase would necessitate more focus on interior finishes in a building LCA.

Due to its influence on product lifetime and emissions, the effects of consumer behavior related to interior finishes needs to be better quantified in order to improve accuracy of residential building LCA. Since lifetime information plays an important role in life cycle studies, and since consumer behavior can greatly influence product lifetime, is it possible to develop models that can accurately predict product lifetime by including both technical factors as well as consumer behavior? Such a tool would not only improve the accuracy of building LCA studies, but also of product comparison studies as well.

Without fully understanding and quantifying the underlying problems, it is not possible to develop effective environmental impact reducing strategies for the built environment. While collecting data for product lifetime, it was noticed that a product's actual lifetime was usually different than what the product was designed for, and was determined by consumer behavior. Therefore, studying the supply chain from the initial design phase down to individual consumer preferences could open new opportunities to reduce the environmental footprint of products and still maintain economy.

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